25901582 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEZGY200, MG4DSEZIM200 - GENERAL TOXICOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Nicotine is classified as a toxicant of: [K] / [CO1]

   a). Animal origin b). Plant origin

   c). Mineral origin d). Microbial origin

2. Xenobiotics with which structural property cross biological membranes most easily by passive diffusion? [U] / [CO3]

   a). High molecular weight b). High lipid solubility

   c). Strong ionic charge d). Large polar surface

3. The compound most frequently causing allergic contact dermatitis in hair dyes is …..... [U] / [CO4]

   a). Resorcinol b). Behentrimonium chloride

   c). PPD d). Cyclomethicone

4. Global economic burden of occupational injuries and diseases is estimated at …... [K] / [CO5]

   a). 3 trillion dollars b). 1 billion dollars

c). 500 million dollars d). 30 trillion dollars

5. Statement 1: Lipophilic compounds like PCBs concentrate in the blubber of marine mammals. Statement 2: Contaminant levels in marine mammals can be thousands of times higher than in the surrounding seawater. [U] / [CO4]

a). Statement 1 is true: Statement 2 is false b). Statement 1 is false; Statement 2 is true

c). Both statements are true d). Both statements are false

6. Which scientist is called the Father of Modern Toxicology? [K] / [CO1]

a). Paracelsus b). Mathieu Orfila

c). Hippocrates d). Alice Hamilton

7. Corpora allata is a gland in insects that produces [K] / [CO3]

a). epinephrin b). juvenile hormone

c). insulin d). acetyl choline

8. Which of the follwing is listed as part of "Supportive Care" for CWA victims? [U] / [CO4]

a). Performing immediate surgery b). Only topical irrigation of wounds

c). Airway, Breathing and Circulation(ABCs), oxygen supplementation and IV access d). Psychological counselling

9. Aflatoxin is mainly produced by _____. [U] / [CO2]

a). Aspergillus flavus b). Rhizopus flavus

c). Candida albicans d). Penicillium notatum

10. DDT is an example of---------- [U] / [CO3]

a). Biodegradable chemicals b). Persistent organic pollutant

c). Water spluble toxin d). Non toxic compound

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

11. What are good laboratory practices in toxicological studies? How is it important? [K] / [CO1]

12. What is the role of Phase I reactions in xenobiotic metabolism? [U] / [CO3]

13. Classify the major classes of Cyanobacterial toxins. [U] / [CO4]

14. What are mechanical hazards? [U] / [CO5]

15. List different proactive engineering measures giving examples [K] / [CO5]

16. What is LD50? [K] / [CO1]

17. How xenobiotics redistributed in the body? [U] / [CO3]

18. Explain Veterinary toxicology. [U] / [CO4]

19. Differeniate flammable and corrosive chemicals with suitable examples [U] / [CO5]

20. How do Agricultural pesticides cause mass mortality in birds? [U] / [CO4]

21. How does lipid solubility influence absorption of Xenobiotics? [U] / [CO3]

22. Explain the concept of Chemical Weapons. [U] / [CO4]

[3x10 = 30]

Part C

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

23. Summarise the A-B-C-D-E classification of adverse drug reactions. [U] / [CO4]

24. Compare byssinosis and bagassosis. [A] / [CO5]

25. Explain sources and bioaccumulation of Mercury. Comment on its acute and chronic toxicity. [U] / [CO3]

26. How do food additives help improve the flavour, colour, texture, and overall appearance of food products? Cite suitable examples. [K] / [CO2]

27. What are essential heavy metals? Explain their role in physiological and biochemical processes [U] / [CO3]

28. Explain the standard drugs that are used to treat nerve agent poisoning. [U] / [CO4]

29. Define occupational cancer and write a short note on the major occupational cancers. [K] / [CO5]

30. Explain the contribution of industrial and environmental pollutants to xenobiotic exposure. [U] / [CO3]

[5x6 = 30]